

Timothy C. Peter
Site Vice President- JAF

JAFP-23-0001
January 10, 2023

United States Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, D.C. 20555-0001

James A. FitzPatrick Nuclear Power Plant
Renewed Facility Operating License No. DPR-059
NRC Docket No. 50-333

Subject: LER: 2021-001-01, Inadequate Protection Devices for DC Motor Field
Shunt Cables Through Separate Fire Areas

Reference: FitzPatrick letter, LER: 2021-001, Inadequate Protection Devices for DC
Motor Field Shunt Cables Through Separate Fire Areas, JAFP-21-0096,
dated October 22, 2021

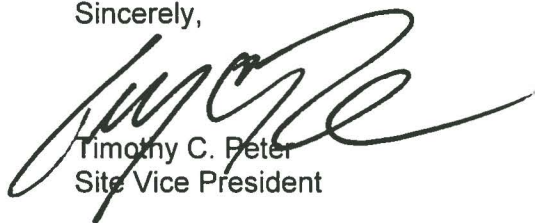
Dear Sir or Madam:

This report is being submitted pursuant to 10 CFR 50.73(a)(2)(ii)(B). This supplement is being submitted to clarify and add more detail to the reported condition and does not change the intent of the original report.

There are no new regulatory commitments contained in this report.

Questions concerning this report may be addressed to Mr. Mark Hawes, Regulatory Assurance Manager (acting), at (315) 349-6659.

Sincerely,



Timothy C. Peter
Site Vice President

TCP/mh

Enclosure: LER: 2021-001-01, Inadequate Protection Devices for DC Motor Field
Shunt Cables Through Separate Fire Areas

cc: USNRC, Region I Administrator
USNRC, Project Manager
USNRC, Resident Inspector
INPO Records Center (IRIS)

NRC FORM 366 (08-2020)		U.S. NUCLEAR REGULATORY COMMISSION		APPROVED BY OMB: NO. 3150-0104 EXPIRES: 08/31/2023	
		LICENSEE EVENT REPORT (LER) (See Page 3 for required number of digits/characters for each block)		Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk all: oira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.	
1. Facility Name James A. FitzPatrick Nuclear Power Plant		2. Docket Number 05000333		3. Page 1 OF 4	
4. Title Inadequate Protection Devices for DC Motor Field Shunt Cables Through Separate Fire Areas					
5. Event Date		6. LER Number		7. Report Date	
Month	Day	Year	Year	Sequential Number	Revision No.
08	24	2021	2021 -	001	- 01
			01	10	2023
				8. Other Facilities Involved	
				Facility Name: N/A Docket Number: N/A	
				Facility Name: N/A Docket Number: N/A	
9. Operating Mode 1			10. Power Level 100		
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)					
☐ 10 CFR Part 20		☐ 20.2203(a)(2)(vi)		☐ 50.36(c)(2)	
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.46(a)(3)(ii)	
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.69(g)	
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.73(a)(2)(i)(A)	
☐ 20.2203(a)(2)(i)		10 CFR Part 21		☐ 50.73(a)(2)(i)(B)	
☐ 20.2203(a)(2)(ii)		☐ 21.2(c)		☐ 50.73(a)(2)(i)(C)	
☐ 20.2203(a)(2)(iii)		10 CFR Part 50		☐ 50.73(a)(2)(ii)(A)	
☐ 20.2203(a)(2)(iv)		☐ 50.36(c)(1)(i)(A)		☒ 50.73(a)(2)(ii)(B)	
☐ 20.2203(a)(2)(v)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(iii)	
☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(v)(A)		☐ 50.73(a)(2)(v)(B)	
☐ 50.73(a)(2)(v)(B)		☐ 50.73(a)(2)(v)(C)		☐ 50.73(a)(2)(v)(D)	
☐ 50.73(a)(2)(v)(D)		☐ 50.73(a)(2)(vii)		☐ 50.73(a)(2)(viii)(A)	
☐ 50.73(a)(2)(viii)(A)		☐ 50.73(a)(2)(viii)(B)		☐ 50.73(a)(2)(ix)(A)	
☐ OTHER (Specify in Abstract below or in NRC Form 366A).					
12. Licensee Contact for this LER					
Licensee Contact Mr. Mark Hawes, Regulatory Assurance Manager (acting)				Telephone Number (Include Area Code) 315-349-6659	
13. Complete One Line for each Component Failure Described in this Report					
Cause	System	Component	Manufacture	Reportable to IRIS	Cause
14. Supplemental Report Expected				15. Expected Submission Date	
☒ No ☐ Yes (If yes, complete 15. Expected Submission date)				Month: Day: Year:	
Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)					
During an extent of condition review for unprotected circuits, the James A. FitzPatrick Nuclear Power Plant (JAF) staff identified three cables for which a postulated fire event could potentially short circuit and cause secondary cable damage. The main circuit breakers in the motor control center compartment do not provide adequate protection of the shunt field circuit cable. This condition affects the DC motors to 27MOV-123, torus exhaust isolation bypass valve, 23MOV-14, HPCI turbine steam inlet isolation valve, and 23MOV-24, HPCI full flow test return to condensate storage isolation valve. The cables for these circuits are routed thru fire areas in South Cable Tunnel, Reactor Building Eastside, and Reactor Building West Crescent; where safe shutdown equipment could be affected by this condition. The cause of the condition is that the original plant design, prior to 10 CFR 50 Appendix R, did not include overcurrent protection for DC motor shunt field cables in all applicable Safe Shutdown Analysis circumstances. Compensatory actions were established for the affected Fire Zones and modifications were completed for the affected circuits.					



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: oira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
James A. FitzPatrick Nuclear Power Plant	05000 – 333	YEAR 2021	SEQUENTIAL NUMBER – 001	REV NO. – 01

NARRATIVE

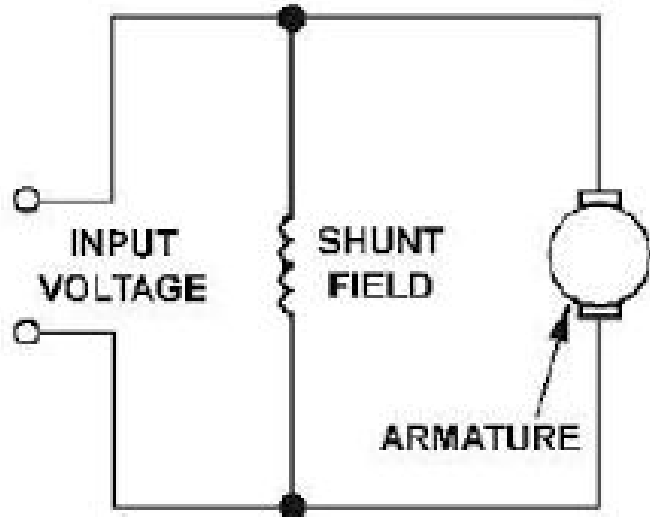
Background

The James A. FitzPatrick Nuclear Power Plant (JAF) 10 CFR 50 Appendix R Safe Shutdown Analysis is based on the occurrence of a single fire. The only failures that are considered are those directly attributable to the fire, and spurious operations that can be postulated to occur as a result of the fire. No other concurrent failures are assumed to occur (i.e., single failure).

Fire areas are established to meet the separation requirements of Section III.G of Appendix R for safe shutdown systems. The fire areas at JAF are further sub-divided into fire zones.

It is postulated that a fire in one fire area could cause a short-circuit, cause overcurrent and overheat cables, then produce secondary damage to adjacent cables in other fire areas where the cables are routed. Fuses in electrical circuits prevent this type of propagation. Secondary cable failures are outside the assumptions of the 10 CFR 50 Appendix R Safe Shutdown Analysis.

In a direct current (DC) shunt motor, the armature and field (shunt) windings are connected in parallel. The shunt (field) windings of a DC shunt motor are made of smaller gauge wire, but they have many more turns than a series-wound DC motor. The high number of turns allows a strong magnetic field to be generated, but the smaller gauge wires provide a high resistance and limit the current flowing through the shunt coil.



Event Description

On August 24, 2021, an extent of condition evaluation from the condition reported by LER: 2020-002, identified three additional circuits, vulnerable to a potential hot short condition impacting secondary fire areas.

The three circuits identified are associated with DC Motors with field shunt cables that transverse multiple fire areas. The feeder breaker protects the armature cables but the smaller 12 AWG (American Wire Gauge) field shunt cables are inadequately protected by the feeder breaker. In the event of a fire-induced hot short, the potential exists, based on the available fault current, cable size, and characteristics of the protective device, for the shunt cable to be overheated.

This situation has the potential to cause secondary cable failures that are outside the assumptions of the 10 CFR 50 Appendix R Safe Shutdown Analysis. This condition was reported to the NRC on August 24, 2021 (ENS 55427) per 10 CFR 50.72(b)(3)(ii)(B). The required actions of the Technical Requirement Manual (TRM) 3.7.M, Fire Barrier Penetrations, were established for the affected fire areas.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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James A. FitzPatrick Nuclear Power Plant	05000 – 333	2021	– 001	– 01

Event Analysis

This condition identified the smaller gauge cables connected to the DC motor shunt field. The cables connected to the armature and series field are of sufficient size to be protected by a 40 or 70 ampere (A) breaker at the Motor Control Center (MCC) and the control circuits are protected by 15 A fuses in series with the breaker.

The 12 AWG shunt field cables were evaluated using the guidance of NUREG-1805 and NUREG/CR-5384. For thermoset insulated cable failures to impact adjacent cables, the temperature that the conductor reaches at given specified current, for a specified time duration, needs to be conducive to this type of damage propagation. At higher fault currents, the DC motor breaker would likely trip prior to cable ignition. However, under some fault current circumstances, the evaluation discovered that these 12 AWG cables are vulnerable to this type of failure. The evaluation further identified that the vulnerable cables were within common enclosures with other equipment cables required for safe shutdown in secondary fire areas.

The following DC motor field shunt cables may impact credited safe shutdown equipment for fires initiated in the following areas:

Cable	Equipment	Noncompliant Fire Area
1PCPARK107	71BMCC-1 to 27MOV-123 Torus exhaust isolation valves 27AOV-117 and 27AOV-118 outbound bypass valve	09 – Reactor Building Eastside 11 – South Cable Tunnel
1HPIBBK006	71BMCC-2 to 23MOV-14 HPCI turbine steam inlet isolation valve	18 – Reactor Building West Crescent
1HPIBBK024	71BMCC-4 to 23MOV-24 HPCI Full Flow test return to condensate storage downstream isolation valve	18 – Reactor Building West Crescent

Since the degree of separation for redundant safe shutdown trains is lacking, the event is reportable as an unanalyzed condition that significantly degraded plant safety. Therefore, this report is being submitted in accordance with 10 CFR 50.73(a)(2)(ii)(B), a condition that resulted in the plant being in an unanalyzed condition that significantly degraded plant safety.

Cause

The overcurrent protection is not sufficient in all applicable DC shunt field cables which can impact the 10 CFR 50 Appendix R Safe Shutdown Analysis. This condition has existed since original plant design.

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CONTINUATION SHEET**

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Similar Events

FitzPatrick, LER: 2013-003-00, Unfused DC Ammeter Circuits Result in Unanalyzed Condition, JAFP-13-0158, dated December 26, 2013.

FitzPatrick, LER: 2020-002-00, Unanalyzed Condition due to Unprotected Control Circuits Running through Multiple Fire Areas, JAFP-20-0032, dated April 14, 2020.

The condition being reported in this report was identified during a Corrective Action Program (CAP) extent of condition review from LER: 2020-002-00. These two LER and the condition reported address unprotected cables which may potentially cause secondary fires such that the safe shutdown analysis is impacted.

Corrective Actions

Compensatory actions for the affected fire areas were established in accordance with TRM 3.7.M. Modifications to add protective devices (i.e, fuses) to the affected circuits was completed in October 2021 (Engineering Change (EC) 634899).

Other DC Motor shunt field cables that were evaluated to not affect safe shutdown analysis are being addressed in the corrective action process (IR 04442380).

An alternating current (AC) extent of condition evaluation concluded that no AC circuit affects the safe shutdown analysis. Circuits evaluated as not fully protected is addressed in CAP.

Safety Significance

Nuclear safety – There were no actual consequences caused by this condition. The potential consequence of a hot short condition is for a fire damage in one fire area to damage other cables in more than one fire area. This may cause a loss of safe shutdown capability outside the analysis of the Safe Shutdown Analysis. The risk of fire damage is mitigated by fire protection equipment and the availability of Fire Brigade members trained to respond to fire accident scenarios.

References

JAF Issue Report IR 04442380, August 24, 2021

JAF-RPT-FPS-01975, Revision 5, 10CFR50 Appendix R Safe Shutdown Analysis Report (SSAR)